# L2. System Modelling - Problem Bank Questions -Suggested Solutions

## L2a. Theoretical Modelling

### Transfer Function

**1.** The linear equation of motion (approximated for small angle about $\theta = 0$) of the single-link pendulum shown in Figure 1 is

$$ml^2\ddot{\theta}(t) + c\dot{\theta}(t) + mgl\theta(t) = u(t)$$

where *m* is the pendulum mass, *c* is the viscous damping, *g* is the gravitational constant, *l* is the distance to the center of mass, and *u* is the applied torque at the rotating base. Find the transfer function $\Theta(s)/U(s)$ of the pendulum assuming the angle starts at zero, $\theta(0^-) = 0$, and is motionless, $\dot{\theta}(0^-) = 0$.

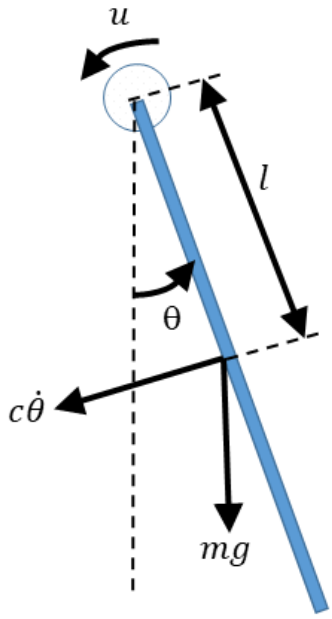



*Figure 1 - Single link pendulum*

**Solution:**

Taking the Laplace Transform of the equation of motion

$$ml^2\left(s^2\Theta(s) - s\theta(0^-) - \dot{\theta}(0^-)\right) + c\left(s\Theta(s) - \theta(0^-)\right) + mgl\Theta(s) = U(s)$$

Given that the initial conditions are zero, $\theta(0^-) = 0$ and $\dot{\theta}(0^-) = 0$,

$$ml^2s^2\Theta(s) + cs\Theta(s) + mgl\Theta(s) = U(s)$$

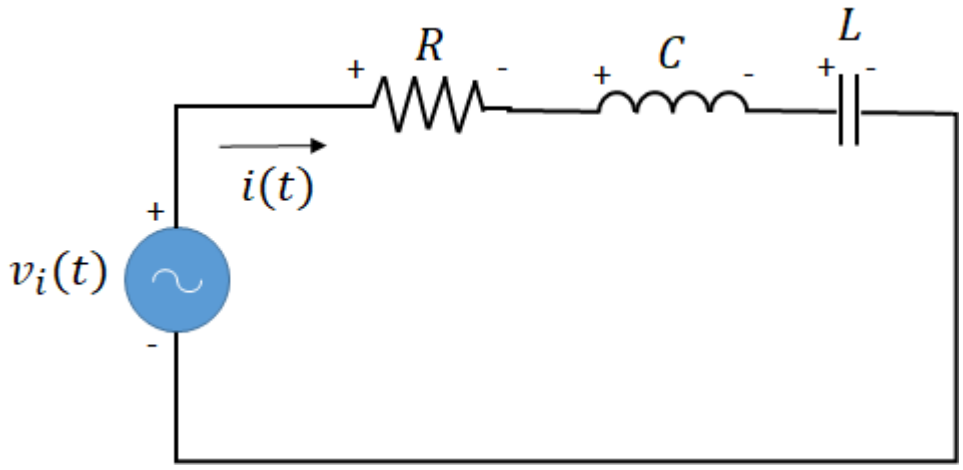
Solving for $\Theta(s)/U(s)$

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$$\frac{\Theta(s)}{U(s)} = \frac{1}{ml^2s^2 + cs + mgl}$$

**2.** The RLC circuit shown below has the following equation.

$$v_i(t) = Ri(t) + \frac{1}{C}\int_0^t i(\tau)d\tau + L\frac{di(t)}{dt}$$




Find the transfer function of the current, $I(s)$, with respect to the input voltage, $V_i(s)$, and for zero initial condition.

**Solution:**

Take the Laplace transform of the circuit equation with zero initials

$$V_i(s) = RI(s) + \frac{1}{Cs}I(s) + LsI(s)$$

Given the initial current is i0 and the grouping the coefficient terms

$$\left(R + \frac{1}{Cs} + Ls\right)I(s) = V_i(s)$$

Solving for $I(s)$

$$I(s) = \frac{1}{R + \frac{1}{Cs} + Ls}V_i(s)$$

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**3.** Solve the following differential equation for a unit step input, $u(t)$, and the initial conditions, $y(0^-) = 1$, and is motionless, $\dot{y}(0^-) = 0$.

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5u(t)$$

**Hint**: Use the partial-fraction expansion to the Laplace transform of the system in order to make it easier to find the inverse Laplace to obtain $y(t)$.

**Solution:**

Take the Laplace transform of the equation

$$s^2Y(s) - s\,y(0^-) - \dot{y}(0^-) + 3\big(sY(s) - y(0^-)\big) + 2Y(s) = \frac{5}{s}$$

Apply the initial conditions

$$s^2Y(s) - s\ + 3sY(s) - 3 + 2Y(s) = \frac{5}{s}$$

Solving for $Y(s)$

$$Y(s) = \frac{s^2\ + 3s + 5}{s(s^2 + 3s + 2)}$$

Factoring the denominator

$$Y(s) = \frac{s^2\ + 3s + 5}{s(s+1)(s+2)}$$

Applying partial-fraction expansion

$$\frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} = \frac{s^2\ + 3s + 5}{s(s+1)(s+2)}$$

To find these terms

$$A(s+1)(s+2) + Bs(s+2) + Cs(s+1) = s^2\ + 3s + 5$$

To find the fraction coefficients: set $s = 0$ to find that $A\ =\ 5/2$; $s = -1$ to find that $B = -3$; set $s = -2$ to find $C = 3/2$. Applying this to the transfer function

$$Y(s) = \frac{5}{2}\frac{1}{s} - \frac{3}{s+1} + \frac{3}{2}\frac{1}{s+2}$$

Take the inverse Laplace

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$$y(t) = \frac{5}{2} - 3e^{-t} + \frac{3}{2}e^{-2t}, \quad \text{for t} \geq 0.$$

## Block Diagrams

**1.** The block diagram in Figure 1 includes two controllers, $C_1(s)$ and $C_2(s)$, to control the plant, $P(s)$. Reduce the block diagram to a single transfer function.

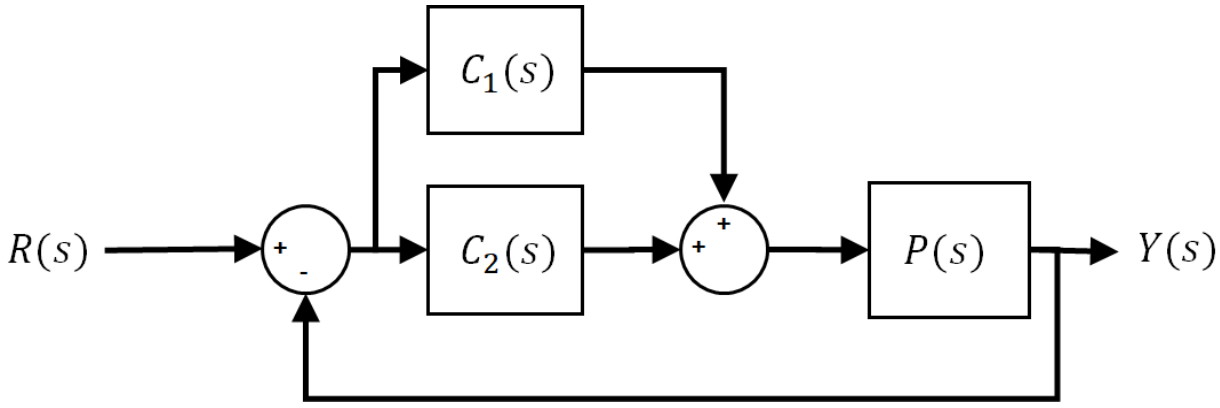
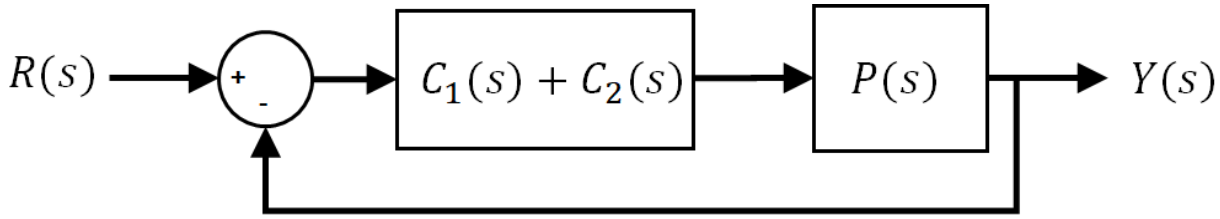


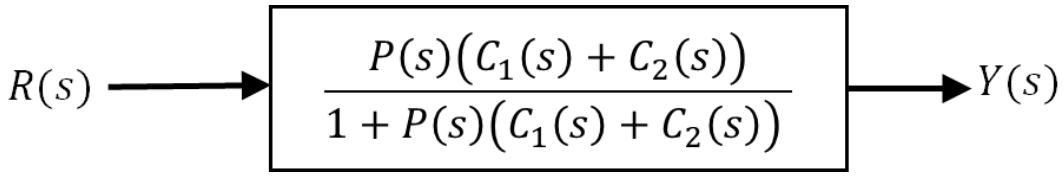

*Figure 1 - Closed-loop system with two controllers*

**Solution:**

The two controller can be added together to result in the following block diagram.




Applying the closed-loop unity feedback block diagram reduction




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**2.** The block diagram in Figure 2 uses multiple control loops to stabilize the plant. Reduce the block diagram to a single transfer function.

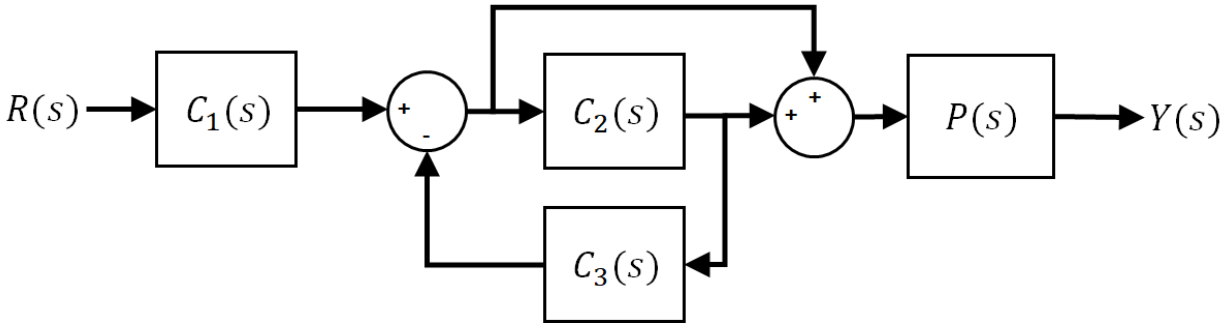
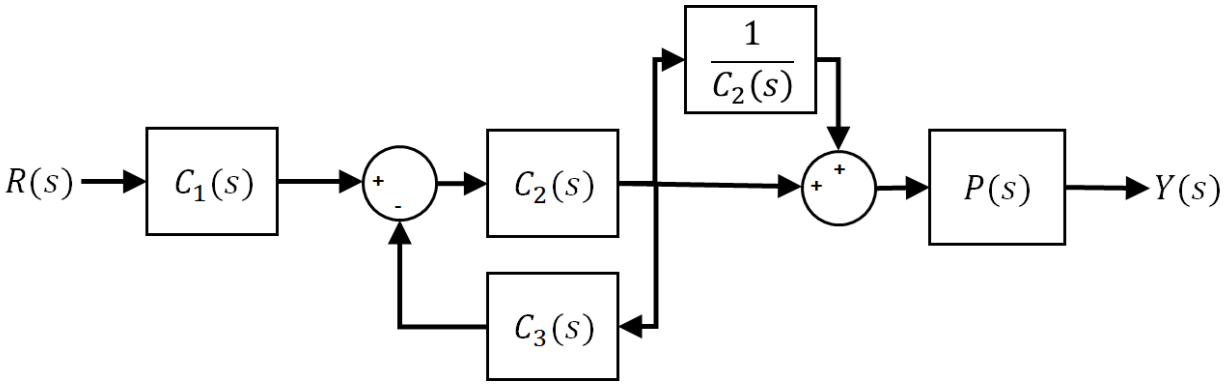


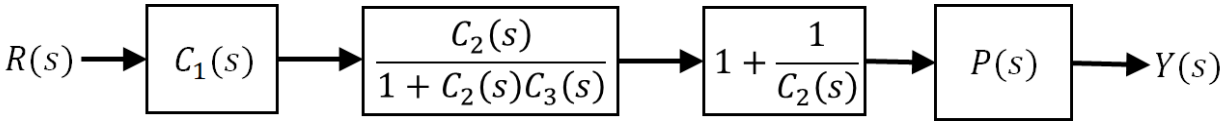

*Figure 2 - Multiple control loops*

**Solution:**

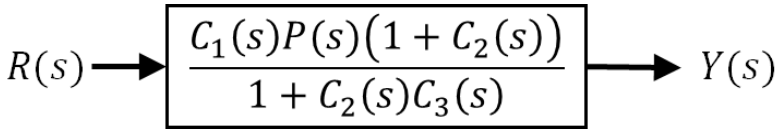
Moving the forward loop after $C_2(s)$ results in the following system.




Reducing the closed-loop system involving $C_2(s)$ and $C_3(s)$ and simplifying the $1/C_2(s)$ summation point results in a series block diagram.




Multiplying all the elements together and simplifying the transfer function.




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## L2b. State Space Modelling

1. Find the linear state-space matrix representation of the two loop circuit given below given that the current in each loop is being measured.

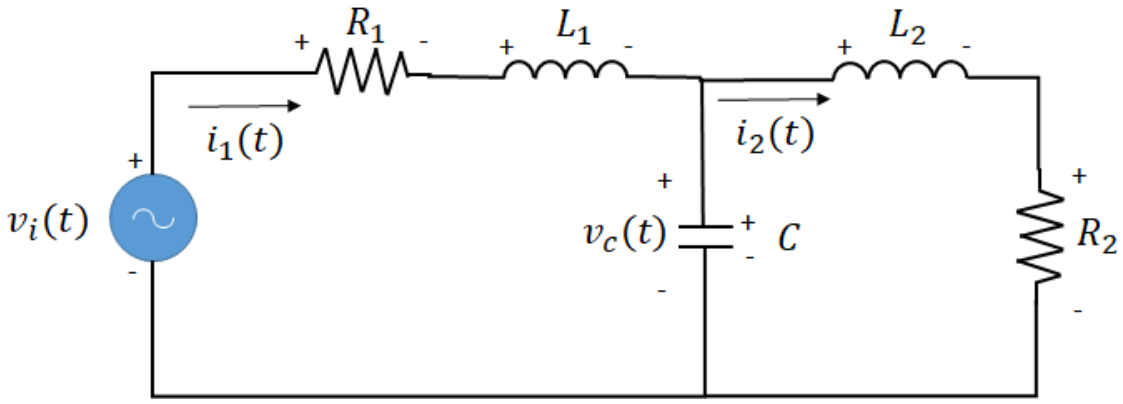



*Figure 1 - Two loop circuit*

The voltage equation for each loop are

$$v_i(t) = R_1 i_1(t) + L_1 \frac{di_1(t)}{dt} + v_c(t)$$

And

$$v_c(t) = L_2 \frac{di_2(t)}{dt} + R_2 i_2(t)$$

The voltage across the capacitor is

$$C \frac{dv_c(t)}{dt} = i_1(t) - i_2(t)$$

**Solution:**

Assigning the following state variables to the loop currents and capacitor voltage

$$x_1 = i_1(t), x_2 = i_2(t), x_3 = v_c(t)$$

and defining the input as $u = v_i(t)$, the circuit equations above can be redefined as

$$u = R_1 x_1 + L_1 \dot{x}_1 + x_3$$

$$x_3 = L_2 \dot{x}_2 + R_2 x_2$$

$$C \dot{x}_3 = x_1 - x_2$$

Solving for the state derivatives

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$$\dot{x}_1 = \frac{1}{L_1}(u - R_1 x_1 - x_3)$$

$$\dot{x}_2 = \frac{1}{L_2}(x_3 - R_2 x_2)$$

$$\dot{x}_3 = \frac{1}{C}(x_1 - x_2)$$

Putting this into the matrix form

$$\dot{x} = \begin{bmatrix} -R_1/L_1 & 0 & -1/L_1 \\ 0 & -R_2/L_2 & 1/L_2 \\ 1/C & -1/C & 0 \end{bmatrix} x + \begin{bmatrix} 1/L_1 \\ 0 \\ 0 \end{bmatrix} u$$

Given that only the loop current are being measured, the output equation is $y_1 = x_1$ and $y_2 = x_2$,

$$y = \begin{bmatrix} 100 \\ 010 \end{bmatrix} x$$

2. The diagram shown in Figure 2 shows the free-body diagram of a harmonic drive. The motor produces a torque of $\tau_m$ and has viscous friction coefficient of $b_m$. Although harmonic drives have many advantages (e.g. zero backlash, high torque), their drawback is they can have nonlinear torsional elements. In this case we model the gear train with as a linear spring with a torsional stiffness *k*.

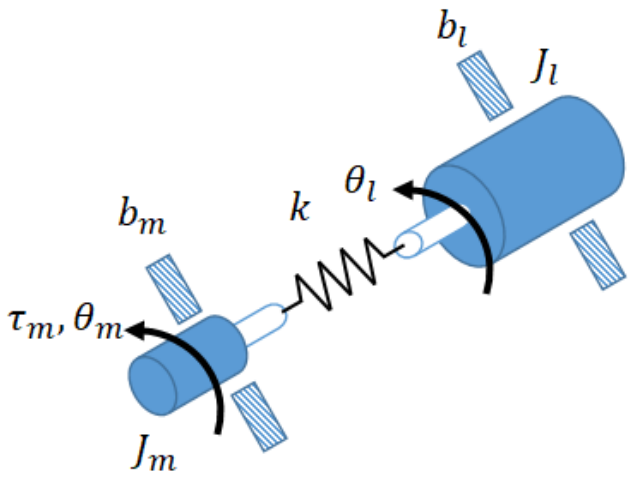



*Figure 2 - Free body diagram of a harmonic drive with torsional flexibilities*

The equation describing the torque input to the motor angle is

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$$J_m\ddot{\theta}_m + b_m\dot{\theta}_m + k(\theta_m - \theta_l) = \tau_m$$

The equation for the load angle is

$$J_l\ddot{\theta}_l + b_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$

Only the difference between the load and motor angels is measured directly, i.e. $\theta_l - \theta_m$. Describe both the state and output equations in matrix form.

**Solution:**

Solving for the acceleration terms

$$\ddot{\theta}_m = \frac{1}{J_m}\left(-k(\theta_m - \theta_l) - b_m\dot{\theta}_m + \tau_m\right)$$

and

$$\ddot{\theta}_l = \frac{1}{J_l}\left(-k(\theta_l - \theta_m) - b_l\dot{\theta}_l\right)$$

Defining the states variables as

$$x_1 = \theta_m, x_2 = \theta_l, x_3 = \dot{\theta}_m, x_4 = \dot{\theta}_l$$

and the input as the applied torque, $u = \tau_m$, the equations become

$$\dot{x}_1 = \frac{1}{J_m}(-k(x_1 - x_2) - b_m x_3 + u)$$

and

$$\dot{x}_2 = \frac{1}{J_l}(-k(x_2 - x_1) - b_l x_4)$$

Given we can only measure the difference between the load and motor angles, the output equation is

$$y = x_2 - x_1$$

The matrix representation of the system is

$$\dot{x} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/J_m & k/J_m & -b_m & 0 \\ k/J_l & -k/J_l & 0 & -b_l \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1/J_m \\ 0 \end{bmatrix} u$$

And

$$y = [-1\ 1\ 0\ 0\ ]x$$

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3. The gantry crane shown in Figure 3 is a prime example of a real-life pendulum-like system. The free-body diagram below shows a force, *u*, being applied to a cart with mass $m_c$. As the cart moves along the support beam, the crane (pendulum) tends to swing back-and-forth. The pendulum has a moment of inertia *J* and the mass $m_p$ and the center of mass from the pivot point is *l*.

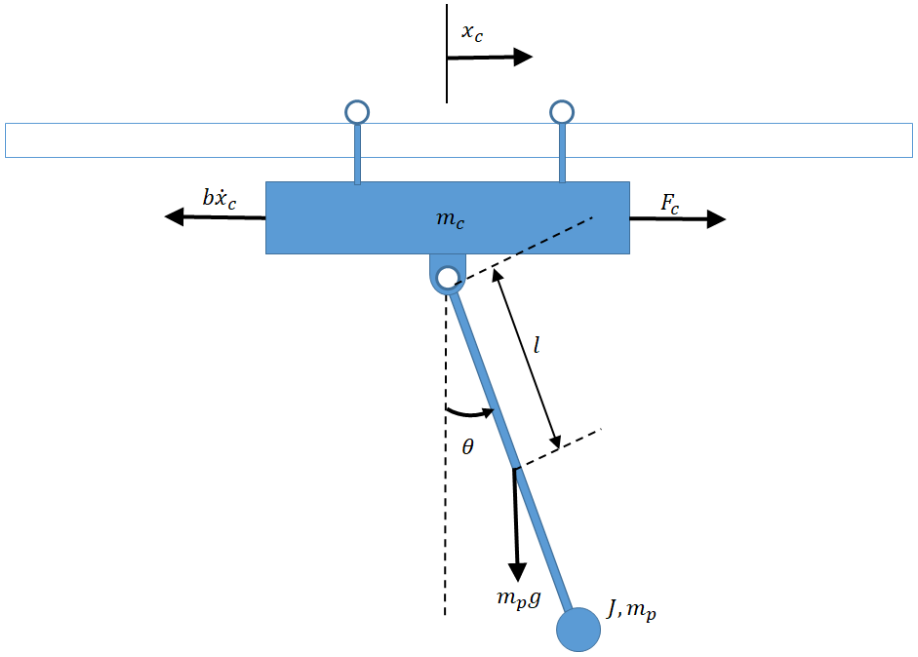



*Figure 3 - Free body diagram of linear gantry carne*

The linearized equations of motion of this system are:

$$m_p l\ddot{x}_c + \left(J + m_p l^2\right)\ddot{\theta} + m_p g l\theta = 0 \quad \text{Eq. 2}$$

And

$$\left(m_c + m_p\right)\ddot{x}_c + b\dot{x}_c + m_p l\theta = F_c \quad \text{Eq. 3}$$

Find the linear state-space representation of the system given that the measured variables are the cart position, $x_c$, and the pendulum/crane angular position, θ, and the control input is the force acting on the cart, $F_c$. Include the A, B, C, D matrix form. Assume no sensor noise.

**Solution:**

We first need to solve for the acceleration terms in the system of equations in Eq. 2 and Eq. 3. Solving for $\ddot{x}_c$ in Eq. 3 $\ddot{x}_c = \frac{1}{m_c+m_p}\left(F_c - b\dot{x}_c - m_p l\theta\right)$

Substituting this into Eq. 2 and solving for the pendulum acceleration

$$\ddot{\theta} = \frac{-m_p g l}{J + m_p l^2}\theta - \frac{m_p l}{\left(J + m_p l^2\right)\left(m_c + m_p\right)}\left(F_c - b\dot{x}_c - m_p l\theta\right)$$

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Collecting like terms

$$\ddot{\theta} = \frac{\left(m_p^2 l^2 - m_p m_c g l - m_p^2 g l\right)\theta + m_p l b \dot{x}_c - m_p l F_c}{\left(J + m_p l^2\right)\left(m_c + m_p\right)}$$

Defining the state as

$$x = \left[x_c \; \dot{x}_c \;\; \theta \; \dot{\theta} \;\right]^T$$

And given the control input is $u = F_c$ the equations of motions become

$$\dot{x_1} = x_2$$

$$\dot{x}_2 = \frac{1}{m_c + m_p}\left(u - b x_2 - m_p l x_3\right)$$

$$\dot{x_3} = x_4$$

$$\dot{x}_4 = \frac{m_p l^2 b}{\left(J+m_p l^2\right)\left(m_c+m_p\right)} x_2 + \frac{\left(m_p^2 l^2 - m_p m_c g l - m_p^2 g l\right)}{\left(J+m_p l^2\right)\left(m_c+m_p\right)} x_3 - \frac{m_p l}{\left(J+m_p l^2\right)\left(m_c+m_p\right)} u$$

Because we are only measuring the cart and pendulum positions, the output equations are

$$y_1 = x_c(t)$$

$$y_2 = \theta(t)$$

The corresponding linear state-space matrix form is therefore

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-b}{m_c + m_p} & \frac{-m_p l}{m_c + m_p} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m_p l^2 b}{\left(J + m_p l^2\right)\left(m_c + m_p\right)} & \frac{\left(m_p^2 l^2 - m_p m_c g l - m_p^2 g l\right)}{\left(J + m_p l^2\right)\left(m_c + m_p\right)} & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{m_c + m_p} \\ 0 \\ -\frac{m_p l}{\left(J + m_p l^2\right)\left(m_c + m_p\right)} \end{bmatrix} u$$

And

$$y = \begin{bmatrix} 1000 \\ 0010 \end{bmatrix} x$$

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## L2c. Linearization

1. The torsional system below has a nonlinear flexible member with the following equation

$$J\ddot{\theta} + b\dot{\theta} + k\,sin\,\theta\ = \tau$$

where *J* is the moment of inertia, *b* is the damping coefficient, and *k* is the stiffness coefficient. Linearize the system for small angles, $\delta\theta$, about $\theta = 0$ and $\dot{\theta} = 0$, and when the applied input torque is zero, $\tau = 0$.

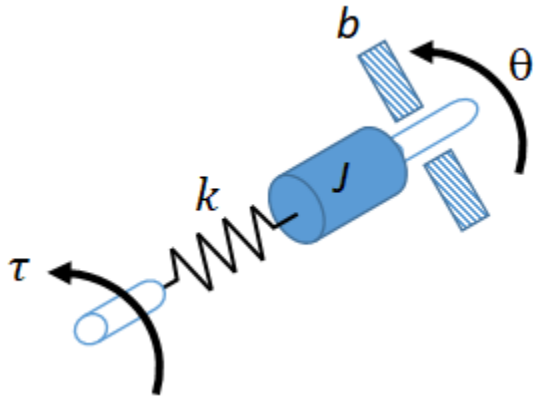



*Figure 1 - Rotary torsional system with a nonlinear flexible coupling.*

**Solution:**

Solving for the acceleration term and defining a three-variable function

$$f(\theta,\dot{\theta},\tau) \coloneqq \ddot{\theta} = -\frac{b}{J}\dot{\theta} - \frac{k}{J}sin\,\theta\ + \tau$$

The general formula for linearizing a multi-variable equation about the operating point $x_0$ is:

$$f_{lin}(x) = f(x_0) + \frac{\partial f(x)}{\partial x_1}]_{x=x_0}(x_1 - x_{10}) + \frac{\partial f(x)}{\partial x_2}]_{x=x_0}(x_2 - x_{20}) + \cdots + \frac{\partial f(x)}{\partial x_n}]_{x=x_0}(x_n - x_{n0})$$

Applying this to the function above given the operational point ($\theta = 0, \dot{\theta} = 0, \tau = 0$):

$$f_{lin}(\theta,\dot{\theta},\tau) = f(\theta_0,\dot{\theta}_0,\tau_0) + \frac{\partial f(\theta)}{\partial \theta}]_{\theta=0\ \dot{\theta}=0\ \tau=0}(\theta - \theta_0) + \frac{\partial f(\theta)}{\partial \dot{\theta}}]_{\theta=0\ \dot{\theta}=0\ \tau=0}(\dot{\theta} - \dot{\theta}_0) + \frac{\partial f(\theta)}{\partial \tau}]_{\theta=0\ \dot{\theta}=0\ \tau=0}(\tau - \tau_0)$$

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Evaluating the expression using the small angle and torque approximations $\delta\theta = \theta - \theta_0$, $\delta\dot{\theta} = \dot{\theta} - \dot{\theta}_0$, and $\delta\tau = \tau - \tau_0$

$$f_{lin}(\theta, \dot{\theta}, \tau) = -\frac{k}{J}\delta\theta - \frac{b}{J}\delta\dot{\theta} + \delta\tau$$

2. The equation describing the heat transfer via radiation is

$$\dot{T}(t) = T(t)^4 + \epsilon T(t) + u(t)$$

where *T* is the temperature, *u* is the input energy, and ε is a thermal coefficient. Find the linearized equation about the operating point $(T_0, u_0)$ where the temperature does not change, i.e. $\dot{T} = 0$. Express this in terms of the small changes about the operating point, $\delta T = T - T_0$ and $\delta u = u - u_0$ .

**Solution:**

The operating point $(T_0, u_0)$ is when the temperature inside the chamber does not change, i.e. $\dot{T} = 0$.

$$0 = T_0^4 + \epsilon T_0 + u_0 \qquad \text{(Eq. A1)}$$

The equation to linearize this about the temperature and energy input variables is:

$$f_{lin}(T, u) = f(T_0, x_0) + \frac{\partial f(T, u)}{\partial T}\Big]_{T=T_0\ u=u_0}(T - T_0) + \frac{\partial f(T, u)}{\partial u}\Big]_{T=T_0\ u=u_0}(u - u_0)$$

Evaluating the equation,

$$f_{lin}(T, u) = f(T_0, u_0) + (4T_0^3 + \epsilon)(T - T_0) + (1)(u - u_0)$$

Given the operating point expression in (Eq. A1) and putting this with respect to the small variable approximation $(\delta x, \delta u)$:

$$f_{lin}(T, u) = (4T_0^3 + \epsilon)\delta T + \delta u$$

The linearized equation is therefore

$$\dot{T} = (4T_0^3 + \epsilon)\delta T + \delta u$$

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3. The Ball and Beam device is a classic control problem of stabilizing the ball to a desired position along the beam. The free-body diagram of a Ball and Beam system that uses a rotary servo to control the angle of the beam is illustrated in Figure 2.

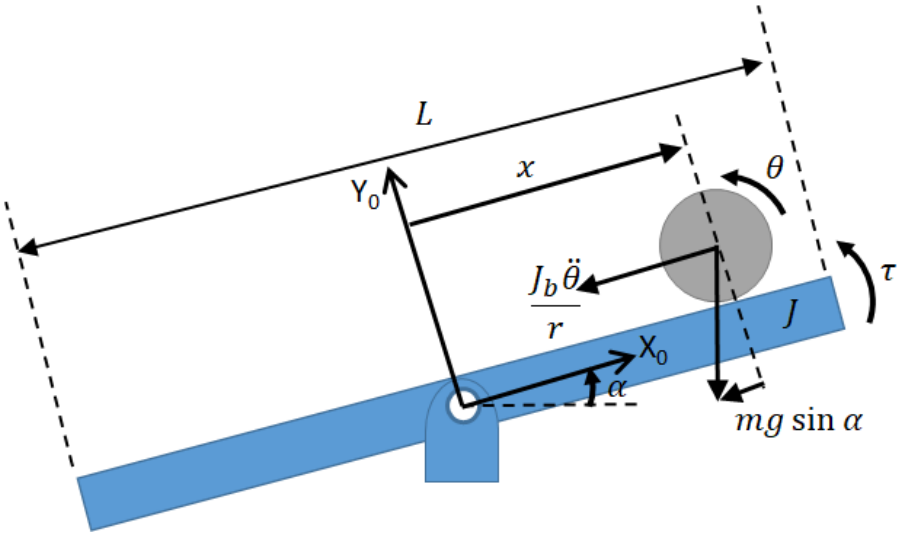



*Figure 2 - Free body diagram of Ball and Beam system*

The nonlinear equations of motions of the system are

$$\frac{7}{5}\ddot{x} + g\ sin\ \alpha\ = 0$$

$$(J + mx^2)\ddot{\alpha} + mg\ x\ cos\ \alpha = \tau$$

where *m* is the mass of the ball, *r* is the radius of the ball, *x* is the linear displacement of the ball along the beam, τ is the applied torque about the beam (the control variable), *L* is the total beam length, α is the angle of the beam relative to the horizontal, and *J* is the moment of inertia of the beam about its rotating axis.

Linearize the system for small angles about the beam horizontal, $\alpha_0 = 0$, and when ball is stabilized in the center of the beam, $x_0 = 0$, and not moving, $\dot{\theta}_0 = 0$.

**Solution:**

At this operating point, the torque equals

$$\tau_0 = (J + mx_0^2)\ddot{\alpha_0} + mgx_0 cos\ \alpha_0$$

At this point, the ball is balanced at $x_0 = 0$, the beam angle is horizontal, $\alpha_0 = 0$, and the ball's acceleration is zero, $\ddot{x} = 0$. Therefore the torque at the operating point is effectively zero,

$$\tau_0 = 0$$

Assigning the following functions to each equation of motion

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$$f_1(\ddot{x},\alpha) := \frac{7}{5}\ddot{x} + g\sin\alpha$$

$$f_2(x,\alpha,\ddot{\alpha},\tau) = (J + mx^2)\ddot{\alpha} + mgx\cos\alpha - \tau$$

Linearizing the first function

$$f_{lin,1}(\ddot{x},\alpha) = f_{lin}(\ddot{x}_0,\alpha_0) + \frac{\partial f(\ddot{x},\alpha)}{\partial \ddot{x}}\Big]_{\ddot{x}=\ddot{x}_0\ \alpha=\alpha_0}(\ddot{x}-\ddot{x}_0) + \frac{\partial f(\ddot{x},\alpha)}{\partial \alpha}\Big]_{\ddot{x}=\ddot{x}_0\ \alpha=\alpha_0}(\alpha-\alpha_0)$$

Resulting in

$$f_{lin,1}(\ddot{x},\alpha) = \frac{7}{5}\delta\ddot{x} + g\delta\alpha$$

Linearizing the 2nd equation based on the variable $x = [x,\alpha,\ddot{\alpha},\tau]$

$$f_{lin,2}(x) = f_{lin}(x_0) + \frac{\partial f(x)}{\partial x}\Big]_{x=x_0}(x-x_0) + \frac{\partial f(x)}{\partial \alpha}\Big]_{x=x_0}(\alpha-\alpha_0) + \frac{\partial f(x)}{\partial \ddot{\alpha}}\Big]_{x=x_0}(\ddot{\alpha}-\ddot{\alpha}_0) + \frac{\partial f(x)}{\partial \tau}\Big]_{x=x_0}(\tau-\tau_0)$$

Evaluating the derivatives

$$f_{lin,2}(x) = (2mx + mg\cos\alpha)\big]_{x=x_0,\alpha=\alpha_0}(x-x_0) - mgx\sin\alpha\,\big]_{x=x_0,\alpha=\alpha_0}(\alpha-\alpha_0) + (J+mx^2)\big]_{x=x_0}(\ddot{\alpha}-\ddot{\alpha}_0) + (-1)\big]\ \ (\tau-\tau_0)$$

At steady state, all the initial conditions are zero. Given this and putting the equation with respect to small variations about the variables

$$f_{lin,2}(x) = mg\delta x + J\delta\ddot{\alpha} - \delta\tau$$

The linearized equation of motions is

$$\frac{7}{5}\delta\ddot{x} + g\delta\alpha = 0$$

$$J\delta\ddot{\alpha} + mg\delta x = \delta\tau$$

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## L2d. Experimental Modelling

1. The step response of a robot arm driving with a DC motor is shown in Figure 1 .

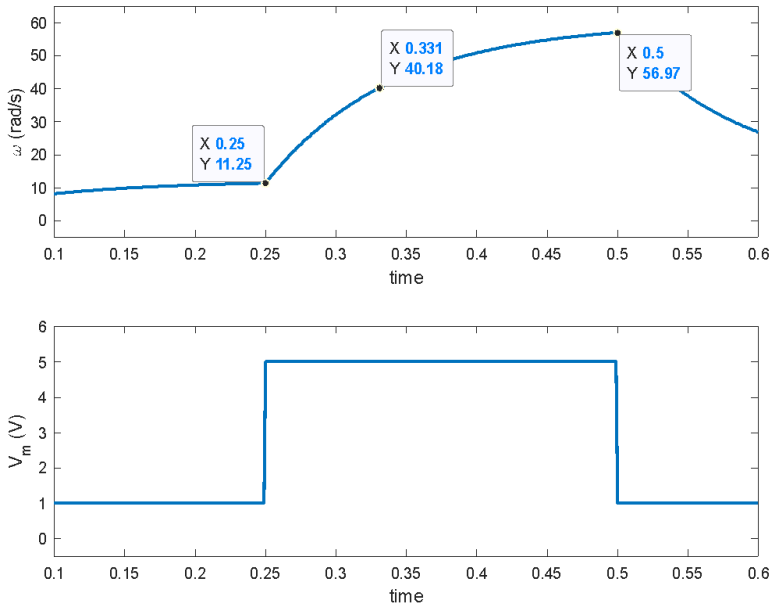



*Figure 1 - Step response of single-link robot arm with DC motor.*

Find the steady-state gain, K, and time constant, τ, parameters of the first-order transfer function

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$$

Use the response data given in Figure 1.

**Solution:**

The steady-state gain is

$$K = \frac{\Delta y}{\Delta u} = \frac{56.97 - 11.25}{5 - 1} = 11.43$$

The time where the speed is 63% (i.e. $1 - e^{-1} = 0.632$) of the maximum angular rate is

$$y(t_1) = (1 - e^{-1})\Delta y + y(t_0)$$

This equals

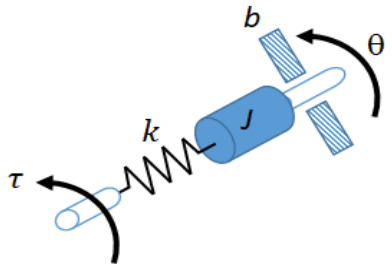
$$y(t_1) = 0.632(56.97 - 11.25) + 11.25 = 40.15$$

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The time at this value is approximately $t_1 = 0.331$ s. Given that the step starts at $t_0 = 0.25$ s, the time constant is

$$\tau = t_1 - t_0 = 0.0801$$

2. Harmonic drives can exhibit flexibilities in their drive. The free-body diagram in is a simplified model of such a system.




The system has an input torque of τ, moment of inertia *J*, a torsional flexibility stiffness of *k*, and a damping coefficient of *b*. The corresponding equation of motion is

$$J\ddot{\theta} + b\dot{\theta} + k\theta = \tau$$

In order to identify the stiffness and damping coefficients, the system is given a small perturbation, i.e. impulse input, and the corresponding angular rate response is measured, as shown in Figure 2.

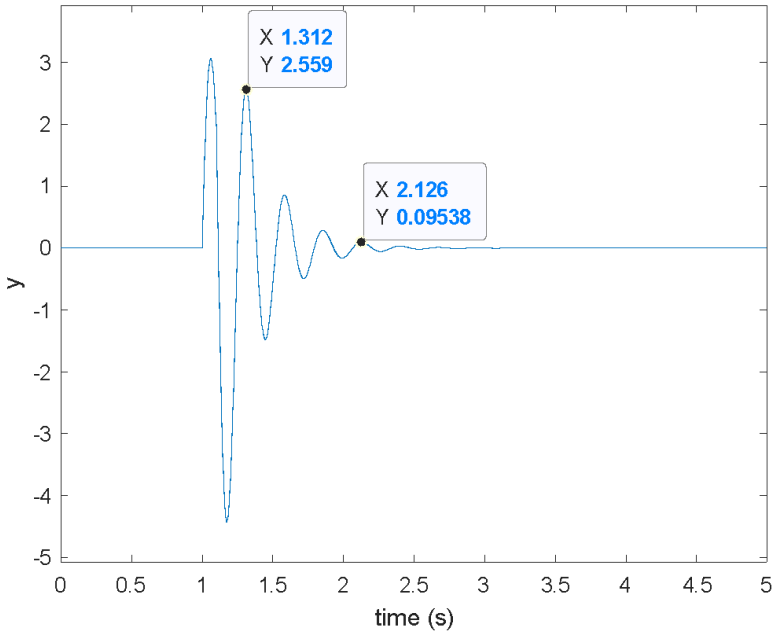



*Figure 2 - Impulse response of harmonic drive*

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Find the stiffness, k, and damping, b, coefficients of the system using the experimentally derived response in Figure 2. The moment of inertia of the system is $J = 0.011$ kg-m$^2$.

**Solution:**

The period of oscillations over $n = 3$ cycles is

$$T_{osc} = \frac{t_5 - t_2}{n} = \frac{2.126 - 1.132}{3} = 0.2713$$

The damped natural frequency is

$$\omega_d = \frac{2\pi}{T_{osc}} = 23.16$$

The logarithmic decrement ratio is

$$\delta = \frac{1}{n} ln \frac{O_2}{O_5} = \frac{1}{3} ln \frac{2.559}{0.09538} = 1.096$$

Given this, the damping ratio is

$$\zeta = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}} = \frac{1.096}{\sqrt{1.096^2 + (2\pi)^2}} = 0.1719$$

Now that the damping ratio is known, we can find the undamped natural frequency

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{23.16}{\sqrt{1 - 0.1719^2}} = 23.51$$

The stiffness and damping coefficients relate to the natural frequency and damping ratio with the following equations:

$$b = 2\zeta\omega_n J$$

And

$$k = J\omega_n^2$$

Given the moment of inertia is 0.011 kg-m$^2$ and the measured values above, the stiffness and damping of the structure are

$$b = 0.0889$$

And

$$k = 6.078$$

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## Frequency Response Modelling

1. The Bode plot of a DC servo motor is shown in Figure 1. Given that this is a first-order system, find the steady-state gain, *K*, cutoff frequency, $\omega_c$, and time constant $\tau$ .

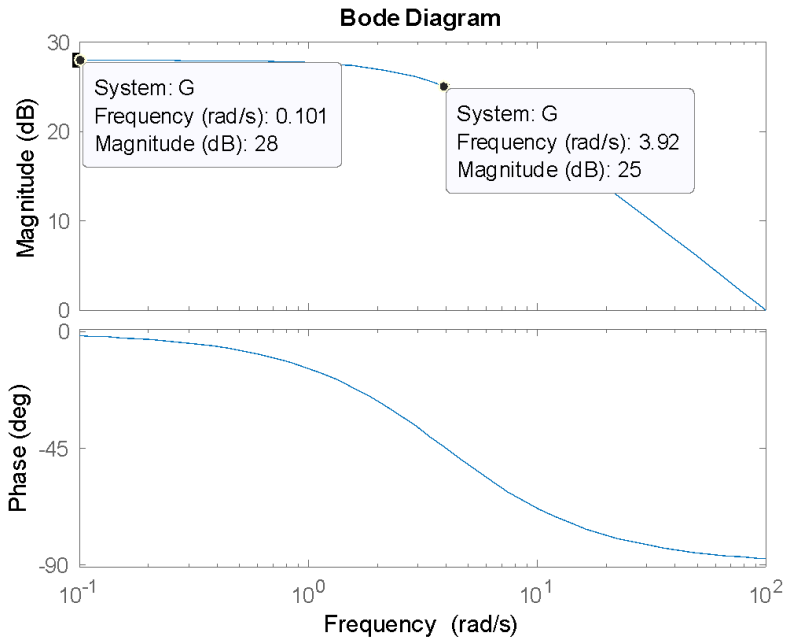



*Figure 1 - Bode plot of DC motor*

**Solution:**

The steady-state gain is the gain at the lowest frequency,

$$K = |G(0.1)| = 28$$

The crossover frequency occurs at the 3 dB gain where $|G(\omega)| = |G(0.1)| - 3 = 25$ dB

$$\omega_c = 3.92$$

The corresponding time constant is therefore

$$\tau = \frac{1}{\omega_c} = 0.255$$

2. Find the transfer function of the first-order system shown in the Bode plot in Figure 3.

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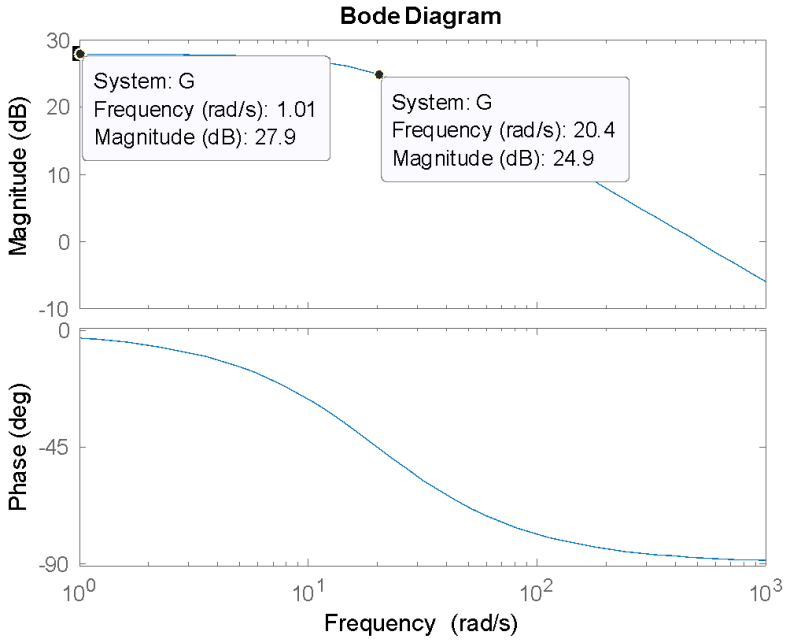



*Figure 3 - Bode plot of first-order system.*

**Solution:**

Given a first-order transfer function of the form

$$G(s) = \frac{K}{\frac{s}{\omega_c} + 1} = \frac{K}{\tau s + 1}$$

The steady-state gain is $|G(0)| = 27.9$ dB which in the linear range is

$$K = 10^{27.9/20} = 24.8$$

The time constant can be found from the crossover frequency

$$\tau = \frac{1}{\omega_c} = \frac{1}{20.4} = 0.049$$

The transfer function is therefore

$$G(s) = \frac{24.8}{0.049s + 1}$$

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